

PASTEL: Progressive Attention-driven Stylization Through Evolving Latents

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Fig. 1. Diverse aesthetic interpretations of the prompt “Starry Night by Van Gogh.” From a single seed, PASTEL enables steerable exploration of stylistic regimes that lie beyond the limits of standard prompting.

Text-to-image diffusion models are constrained by the limits of language, often leaving aesthetic regimes like complex brushwork or emergent textures beyond the reach of textual control. We present PASTEL, Progressive Attention-driven Stylization Through Evolving Latents (<https://jessysu.github.io/PASTEL/>), a framework that treats the attention pathways of pretrained latent diffusion models as a navigable style space. Inspired by advances in attention-driven conditioning [2], PASTEL provides a principled mechanism to systematically curate these expressive possibilities. By modulating cross- and self-attention blocks with configuration parameters during denoising, our approach facilitates a continuum of stylizations, enabling the exploration of latent aesthetic potential.

1 Introduction

Generative diffusion models have demonstrated a remarkable capacity for synthesis, yet they are inherently constrained by the limits of language. The process of artistic creation often demands more than the output of a textual prompt, as it requires curating specific aesthetics that are “unpromptable,” or emergent stylistic qualities that remain ineffable until they are realized. For instance, a model cannot synthesize a specific brushwork style like Van Gogh’s *Starry Night* without explicit prior training, as the conceptual sketch of such styles cannot be fully articulated through language alone.

While early Neural Style Transfer established style as a separation of content from deep feature statistics [1], and StyleGAN later revealed that latent vectors encode disentangled stylistic attributes [3], recent latent diffusion models [4] have shifted the focus toward attention mechanisms. PASTEL leverages this by treating attention pathways as a multi-dimensional style space. In a standard U-Net backbone, self-attention governs structural coherence while cross-attention injects semantic conditioning. The framework intervenes by selectively muting pathways and applying scaling rules that mimic how an artist employs emphasis, abstraction, and compositional balance to shape the expressive intent of a work, allowing the user to guide the evolution of the latent representation. This shifts the paradigm from a text-dependent generator to an exploratory tool, allowing users to curate interpretations that originate from the model’s internal potential rather than the constraints of the input prompt.

2 The PASTEL Framework

PASTEL treats the internal attention blocks of a frozen U-Net as a navigable stylistic space. Instead of uniform attention application, we systematically modulate these pathways to shape the image’s aesthetic evolution.

2.1 Iterative Ranked Modulation

We define the tunable subset B of all attention blocks A in the U-Net. At the start of denoising ($t = 0$), we mute B to isolate the latent representation from standard stylistic priors. We then progressively re-enable these blocks through an iterative Rank-and-Update loop:

- **Sensitivity Ranking:** At each denoising step, we test the impact of marginally increasing each blocked layer’s attention. We measure the resulting deviation Δ_b in the predicted latent.
- **Selective Activation:** Using a policy π , we select a subset k of blocks. Policy $\pi_1 \in \{most, least\}$ determines the ranking priority based on influence magnitude, while $\pi_2 \in \{abs, rel\}$ specifies the aggregation mode, measuring either absolute deviation (*abs*) or directional influence (*rel*).
- **Evolutionary Step:** The selected blocks are increased by a step size s . This allows the latent to "evolve" toward the prompt’s semantics while retaining the unique abstractions generated by the specific activation order.

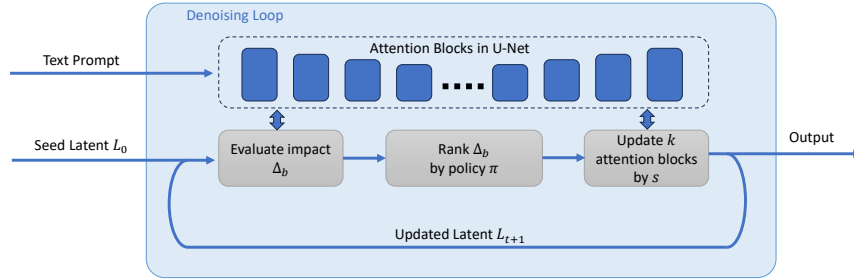


Fig. 2. The PASTEL Loop: A dynamic ranking mechanism that selects attention blocks to update at each denoising step based on their immediate influence on the latent image.

2.2 Navigating the Style Space

A PASTEL *configuration* comprises the tuple $[\pi, B, s, k]$. We frame these parameters as control axes for creative discovery:

- **Policy (π):** Sets the creative direction. *Abs* modes prioritize semantic fidelity, while *rel* modes favor expressive, abstract deviations. Yet *most*-influential ranking encourages high compositional richness whereas *least*-influential promotes consistency and stability.
- **Preset (B):** Defines stylistic scope, by isolating interventions to spatial domains—targeting global structural coherence (encoder blocks) or fine-grained textural detail (decoder blocks).
- **Intensity (s, k):** Governs intervention pace. Lower values preserve abstraction, while higher values accelerate prompt alignment. Aggressive settings (e.g., larger s or more k) can cause summed attention to exceed 1.0, triggering "over-saturation," a regime where attention spikes reveal volatile, novel aesthetic forms.

By varying these parameters, we transform the diffusion process from a "black box" into a structured, traversable landscape of aesthetic possibilities (Fig. 3).

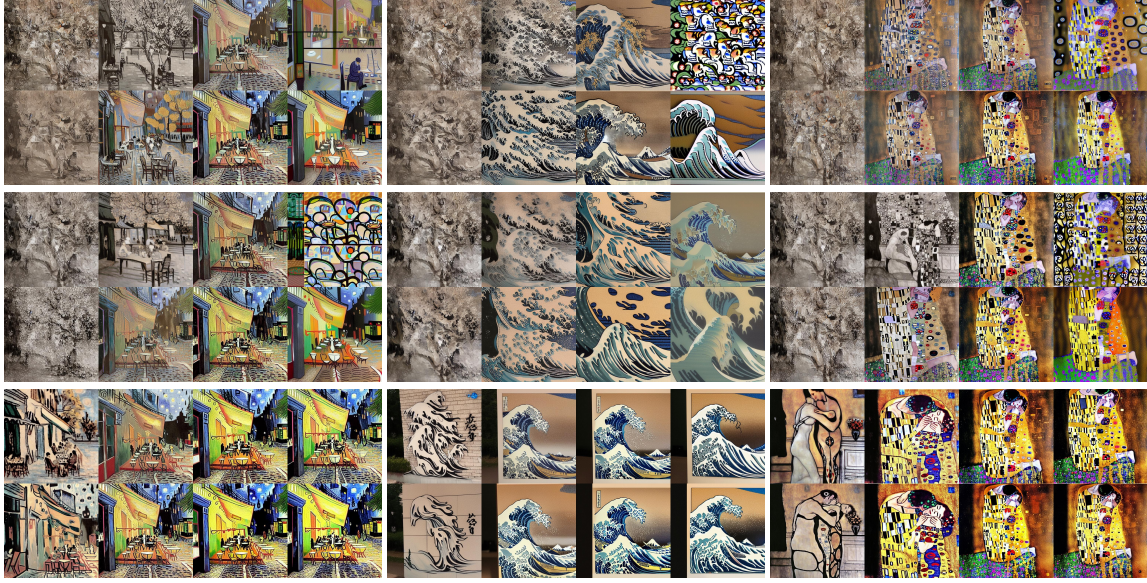


Fig. 3. Concept stylization from seed 0. Rows represent $[\pi, B]$ configurations: $[\pi_{most}, B_{all}]$, $[\pi_{least}, B_{all}]$, and $[\pi_{least}, B_{attn2}]$. Grid columns: $s \in \{0.2, 0.5, 1, 1.25\}$ (left-to-right); grid rows: $k \in \{2, 3\}$ (top-to-bottom). Cumulative attention gain triggers over-saturation novelty when the accumulated attention exceeds 1.0, in the rightmost columns ($s = 1.25$).

3 Results and Exploration

We analyze PASTEL across 720 configurations, producing 68,120 images. We frame the results not as a search for a single "best" image, but as the exploration of a **Perspective Space** where users navigate trade-offs between semantic adherence and abstraction.

3.1 Quantifying the Style Space

We use three metrics as navigational proxies: **Prompt Relevancy** (CLIP alignment), **Aesthetic Score** (LAION quality), and **Image Relevancy** (deviation from baseline).

As shown in Fig. 4, the modulation policy π dictates the regime of exploration. In the left plot, the *abs* policy (blue) favors representational fidelity and higher prompt relevancy. In contrast, the *rel* policy (red) exposes a high-aesthetic, low-relevancy regime, producing abstract interpretations distinct from the baseline. The right plot illustrates the trade-off between prompt adherence and baseline deviation; PASTEL configurations allow for high semantic alignment even when significantly departing from the original image’s structure, providing a wide novelty range for artistic selection.

3.2 User-Centric Curation

To manage the density of the stylization space, we introduce a *Target Score* that balances novelty (low image relevancy) with perceptual quality and semantic grounding. To assist artistic curation, we apply **Farthest Point Sampling (FPS)** in CLIP embedding space to diversify outputs. This ensures the user is presented with a broad stylistic palette rather than redundant variations, transforming PASTEL into a workflow for collaborative aesthetic discovery (Fig. 5).

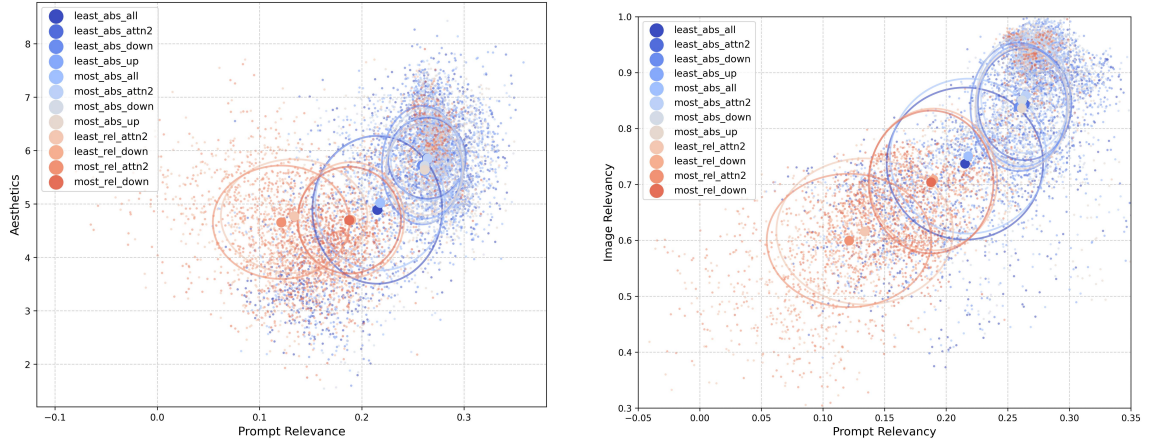


Fig. 4. Quantitative distribution of the stylization space. The *abs* policy (blue) prioritizes representational alignment, while *rel* (red) drives high-aesthetic departures from the default output. These deviations characterize “baseline novelty,” where the model explores stylistic interpretations that remain semantically grounded despite significant structural variance from the unmodified diffusion path.

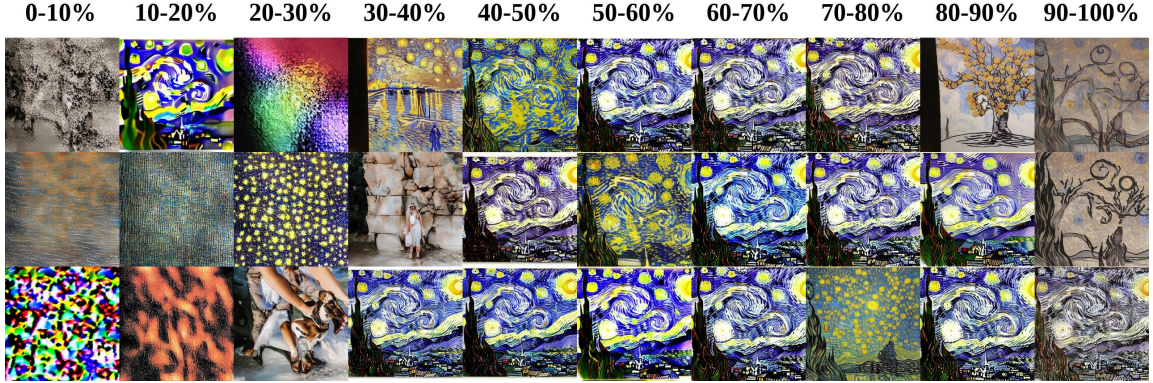


Fig. 5. Style stratification for “Starry Night.” Grouping by aesthetic bins allows for structured navigation of the continuum from abstract noise to representational fidelity.

4 Conclusion

PASTEL introduces a principled approach to traversing the latent stylization spaces of diffusion models. By shifting from a single-output pipeline to a structured, navigable perspective space, we provide a mechanism for fine-grained aesthetic control. Our results demonstrate that policies, presets, and attention scales act as essential navigational axes for artistic exploration, enabling a workflow where users curate transitions between literal representation and expressive abstraction. Future work will integrate these controls into a real-time interactive interface to further close the gap between artistic intent and algorithmic execution.

References

- [1] Leon A Gatys, Alexander S Ecker, and Matthias Bethge. 2016. A Neural Algorithm of Artistic Style. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*.
- [2] Amir Hertz, Naomi Peri, Rinon Gal, Gal Chechik, and Daniel Cohen-Or. 2023. Prompt-to-Prompt Image Editing with Cross-Attention Control. In *International Conference on Learning Representations (ICLR)*.
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- [4] Robin Rombach, Andreas Blattmann, Daniel Lorenz, Patrick Esser, and Björn Ommer. 2022. High-Resolution Image Synthesis with Latent Diffusion Models. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 10684–10695.

A Extended Theoretical Framework and Motivation

A.1 Representational vs. Abstract Art Continuum

The representational tradition in art aims to depict recognizable subjects from the physical world, allowing viewers to identify objects, scenes, and figures. In contrast, abstract art departs from literal appearance and instead emphasizes formal elements such as shape, color, rhythm, and composition. Although the underlying subject may remain faintly discernible, abstraction foregrounds interpretation and stylization over direct representation. This continuum—from representational fidelity to abstraction—has long served as a lens for understanding artistic intent and viewer perception, and provides a natural framework for examining the generative behavior of diffusion models under systematic manipulation.

Recent text-to-image diffusion models have demonstrated remarkable capacity to synthesize visually compelling and semantically aligned images. However, most generation pipelines implicitly assume a single optimal output per prompt, often guided by fixed objective functions or learned aesthetic priors. Such formulations obscure the fact that aesthetic preference is inherently subjective: different viewers may value semantic fidelity, abstraction, texture, or novelty to varying degrees. As a result, the space of plausible and artistically meaningful stylizations remains largely underexplored, and mechanisms for navigating this space in a structured and interpretable way are limited.

In a latent diffusion model, the U-Net backbone iteratively denoises a latent representation, with cross-attention and self-attention layers playing complementary roles. Cross-attention injects information from the text prompt at every step, guiding spatial features toward semantic alignment, while self-attention governs internal feature interactions within the latent, shaping structure, coherence, and style independent of the prompt. When all attention pathways operate normally, these signals are jointly integrated to produce a coherent, representational output consistent with the prompt.

PASTEL modifies this process by selectively controlling how attention layers contribute during denoising. Conceptually, this parallels an artist who retains the subject’s essence while progressively emphasizing expressive formal elements. Our baseline intervention mutes all 32 attention blocks (both cross- and self-attention) in the Stable Diffusion 1.5 U-Net. Using a 40-step denoising schedule, we then progressively re-enable attention: at each iteration, we activate the muted block whose activation would yield the largest immediate change in the latent if applied at that moment.

When all attention is disabled throughout denoising, no prompt information can shape the latent; the model converges to an image determined solely by the noise seed. This outcome is qualitatively distinct from supplying an empty prompt while keeping attention active, which still yields a coherent but prompt-agnostic concept. As progressively more attention blocks are re-enabled, the outputs begin to selectively capture certain prompt elements while suppressing others, reflecting the order and sensitivity with which different attention modules respond to textual conditioning.

B Extended Quantitative Analysis and Canonical Grounding

B.1 Parameter Sensitivity Breakdown

To further characterize the stylization space, we analyze metrics across the number of picks (k) and step size (s).

Table 1. Metrics over number of picks per iteration (k)

k	prompt rel.	image rel.	aesthetics	complexity
1	0.196	0.712	5.059	1.013
2	0.211	0.744	5.223	0.992
3	0.220	0.764	5.298	0.977
5	0.226	0.781	5.322	0.960
7	0.230	0.791	5.323	0.953
10	0.234	0.801	5.352	0.942

Table 2. Metrics over step size (s), sorted by aesthetics

s	prompt rel.	image rel.	aesthetics	complexity
0.5	0.230	0.800	5.559	0.985
0.34	0.220	0.772	5.534	0.998
0.25	0.212	0.753	5.487	1.007
0.2	0.206	0.740	5.470	1.013
1	0.232	0.818	5.456	0.965
0.4	0.224	0.773	5.442	0.987
0.6	0.231	0.785	5.403	0.972
1.25	0.228	0.777	5.163	0.946
0.75	0.208	0.721	4.624	0.942
1.5	0.205	0.716	4.490	0.916

B.2 Canonical Artwork Grounding

To contextualize the behavior of our automated selection metric:

$$\text{Target Score} = \frac{\text{aesthetics} \times \text{prompt_relevancy}}{1 + \text{image_relevancy}} \quad (1)$$

we evaluate the original *Starry Night* by Van Gogh using the same metrics applied to PASTEL outputs. The original painting attains an aesthetic score of 6.17 (82.78th percentile), complexity 1.07 (75.56th percentile), and an image-relevancy baseline of 0.90 (49.72th percentile). Its prompt relevancy with respect to “Starry Night by Van Gogh” is 0.27, placing it in the 83.19th percentile. Together, these values yield a target score of 0.88, ranking in the 90.56th percentile across all generated samples.

Among the 720 PASTEL configurations, 124 produce higher aesthetic scores than the original painting, and 121 exceed its prompt relevancy. This comparison highlights that while progressive stylization can occasionally surpass a canonical artwork along individual dimensions such as aesthetic appeal or semantic alignment, the original occupies a high joint aesthetic–semantic position. Rather than serving as an optimization target, the canonical artwork functions as an anchoring reference within the perspective space defined by progressive stylization.

C Prompt List for Experimental Evaluation

- P1 “The Girl with a Pearl Earring by Vermeer”
- P2 “Starry Night by Van Gogh”
- P3 “Persistence of Memory by Dali”
- P4 “The Scream by Munch”
- P5 “Water Lily Pond by Monet”
- P6 “The Great Wave off Kanagawa”
- P7 “The Kiss by Klimt”

P8 “A desert made of broken mirrors reflecting an endless sunset, ultra detailed, ethereal vibe”

P9 “A city floating above clouds made of cotton candy, dreamy pastel lighting, cinematic depth of field”

P10 “A photo of an astronaut riding a horse on mars”

P11 “A portrait in the style of Rembrandt but with cyberpunk neon reflections”

P12 “Coastal lighthouse during a storm, dramatic sky, long exposure realism”